P802.1Qdq/D2.0
November 25, 2025
(Amendment to IEEE Std 802.1Q™–2022 as amended by IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023, IEEE Std 802.Qcj-2023, IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024, and IEEE Std 802.1Qdy-2025)

Draft Standard for Local and metropolitan area networks—

Bridges and Bridged Networks

Amendment 41: Shaper Parameter Settings for Bursty Traffic Requiring Bounded Latency

Prepared by the
Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

Sponsor
LAN/MAN Standards Committee
of the
IEEE Computer Society

This and the following cover pages are not part of the draft. They provide revision and other information for IEEE 802.1 Working Group members and participants in the IEEE Standards Association ballot process, and will be updated as convenient. The text proper of this draft begins with the page 4.

Current draft

The text of this draft, following these cover pages, is that of P802.1Qdq as of the final Working Group Recirculation Ballot with the omission of change bars showing changes from prior drafts.

Base standard

This draft is being submitted to SA Ballot prior to completion of the SA Ballot of the updated base standard, IEEE Std 802.1Q. There is currently no technical overlap between these amendments, and the editorial interaction is limited to the numbering of minor clauses and figures which are added to the end of their enclosing clause. The current numbering in this draft anticipates the prior approval and pre-publication editing of P802.1Qdq, while the editing instructions accommodate the possibility of further numbering changes.

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Abstract: This amendment to IEEE Std 802.1Q-2022 as amended by IEEE Std 802.1Qcz-2023, IEEE Std 802.1Qcw-2023, IEEE Std 802.1Qcj-2023, IEEE Std 802.1Qdj-2024, IEEE Std 802.1Qdx-2024, and IEEE Std 802.1Qdy-2025 provides recommendations for traffic shaper parameters that reduce the bandwidth provisioning required to support timely and reliable delivery of frames transmitted by devices that generate bursty traffic and frames transmitted by other devices that use the same bridged network.

Keywords: amendment, Bridged Local Area Network, IEEE 802.1Q™, IEEE 802.1Qdq™, LAN, Local Area Network, MAC Bridge, traffic shaping, Virtual LAN, VLAN.

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Jessy Royer, *Vice Chair*
János Farkas, *Time-Sensitive Networking Task Group Chair*
Hiroki Nakano, *Editor*

The following members of the individual balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

A.N. Other

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Introduction

This introduction is not part of IEEE Std 802.1Qdq™-2025,
IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks—
Amendment <tbd>: Shaper Parameter Settings for Bursty Traffic Requiring Bounded Latency

This amendment to IEEE Std 802.Q™-2022 adds an informative annex that describes traffic shaper parameter settings for bursty traffic requiring bounded latency. Appropriate settings for these parameters reduces the need to over-provision the network to mitigate frame loss for bursty traffic, and reduces the transit delay and jitter impacts of frames of other flows.

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IEEE Standard for
Local and metropolitan area networks—

Bridges and Bridged Networks

Amendment: Shaper Parameter Settings for Bursty Traffic Requiring Bounded Latency

[This amendment is based on IEEE Std 802.1Q™-2022 as amended by ...]

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italics***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strikethrough~~ (to remove old material) and <u>underscore</u> (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this note will not be carried over into future editions because the changes will be incorporated into the base standard.

Insert the following new annex:

Annex X

(informative)

Shaper Parameter Settings for Bursty Traffic Requiring Bounded Latency

X.1 Introduction

This annex provides recommended shaper parameter settings for a type of traffic requiring bounded latency that occurs sporadically and that consists of multiple frames. This type of traffic is discussed in IEEE-SA Industry Connections [B1] where Tables 2 to 14 list applications in conjunction with their respective communication requirements. Some of these applications require delivering more than 10,000 bytes of data within an order of 100 milliseconds for monitoring and human safety purposes. Such traffic is an example of bursty traffic requiring bounded latency.

IEEE Std 802.1Q time-sensitive bridged network equipped with any traffic shaper or gate control is capable of handling this type of traffic and helping to provide the bounded latency. Figure X-1 shows an example of the network configuration under consideration. This network comprises Talker X, Talker Y, Listener X, Listener Y, Bridge 1, Bridge 2, and Bridge 3, which connect directly or indirectly to each other. Talker X and Talker Y each generate a reserved stream (Stream X and Stream Y, respectively) that flows via Bridge 1, Bridge 2, and Bridge 3 to a corresponding Listener (Listener X and Listener Y, respectively).

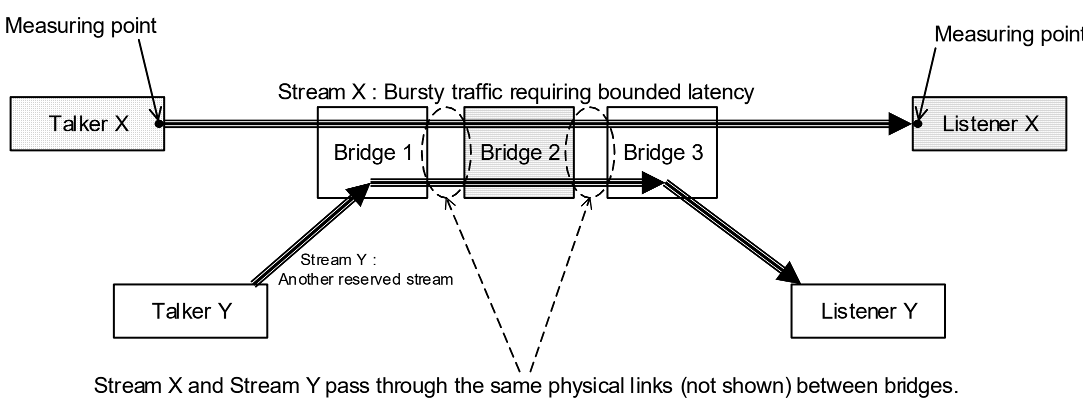

Figure X-1—An example of network structure under consideration

In this example, Stream X and Stream Y flow through common links and bridges in the network. Traffic shaping is performed in the Talkers and resource reservations are performed in bridges based upon the traffic characteristics of the streams. This enables these streams to be transported as planned. IEEE Std 802.1Q specifies multiple transmission selection algorithms for the purpose of traffic shaping, for instance, the credit-based shaper transmission selection algorithm (8.6.8.2) and/or the ATS transmission selection algorithm (8.6.8.5). Any of the transmission selection algorithms, including any combination of the transmission selection algorithms, can be used for this purpose.

Figure X-2 shows the functional structure of a Talker. A Talker can have only a data source and an embedded shaper as transmission selection.

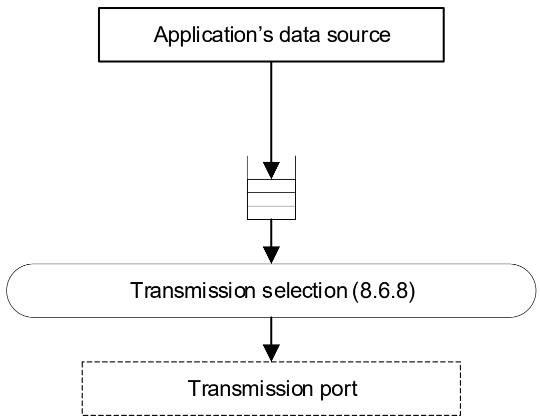

Figure X-2—Functional structure of Talker

The regulation of the outbound traffic in a Talker helps achieving bounded latency.

X.2 Bursty Traffic Requiring Bounded Latency

This clause discusses bursty traffic requiring bounded latency and the parameters that describe it. These parameters comprise times expressed in seconds and otherwise data quantities expressed in bits.

Figure X-3 illustrates the bursty traffic pattern at the egress port of the Talker emitting a single type of traffic without any traffic shaping. A frame cluster is a sequence of contiguous frames that the application considers as a single transaction. B_i denotes the total size of the i-th frame cluster in bits belonging to the i-th transaction of the application and B_{max} denotes the upper bound of B_i for all i. Bounded latency, hereinafter referred to as $L_{bounded}$ is a primary parameter of this traffic and is assumed to be pre-determined by an application or set manually by an operator of an application. $L_{bounded}$ is the maximum transaction delay, that is, measured from the time that a known reference bit in the first frame of a frame cluster passed a known point in Talker to the time that a reference bit in the last frame of the frame cluster passes a known point in Listener. In view of the characteristics of some data transmission with a large interval between frame clusters that can exceed several tens of milliseconds or of event-driven data generation by internet of things devices (see IEEE-SA Industry Connections [B1]), the traffic treated here is sporadic, with condition that the next frame cluster does not arrives until the entire corresponding queue in the sender becomes empty. This condition is satisfied when $L_{bounded}<C_{min}$, where C_{min} denotes the lower bound of C_i for all i, C_i denoting the time between the beginning of the i-th frame cluster and the beginning of the (i+1)-th frame cluster.

In this annex, bursty traffic is characterized by the three given parameters; B_{max}, $L_{bounded}$, *and* C_{min}.

Figure X-4 illustrates the detailed traffic pattern and queues of bursty traffic requiring bounded latency from the application's point of view. At the time t_i, a transmitting application starts the i-th transaction whose size is B_i that is equal to or less than B_{max} and is greater than a frame the bridged network can handle. Therefore, the data of the transaction is carried as a frame cluster comprising n_i frames whose indices k are from 1 to n_i and whose sizes are $F_{i,k}$ respectively, and each frame is transmitted in order. $F_{i,k}$ comprises a part of the i-th

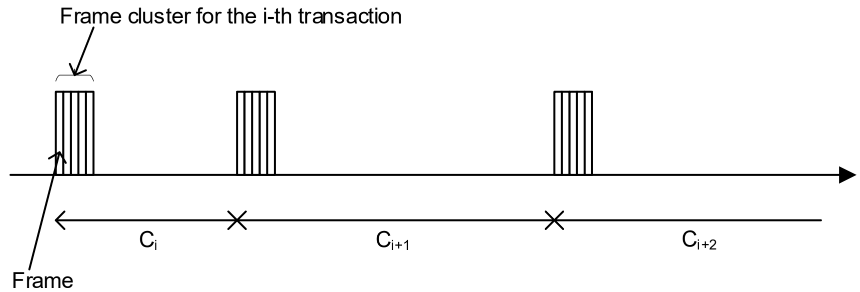

Figure X-3—Bursty traffic pattern of frame cluster

frame cluster and overhead, for instance, frame headers as specified in IEEE 802.1, 802.3, or 802.11 standards. A known reference point in the last frame of the frame cluster needs to reach the corresponding receiving application through the bridged network by the time t'_i that is equal to or less than t_i plus $L_{bounded}$. In addition, the transmitting application starts transmitting the subsequent transaction at time t_{i+1}, which should be equal to or greater than t_i plus C_{min}.

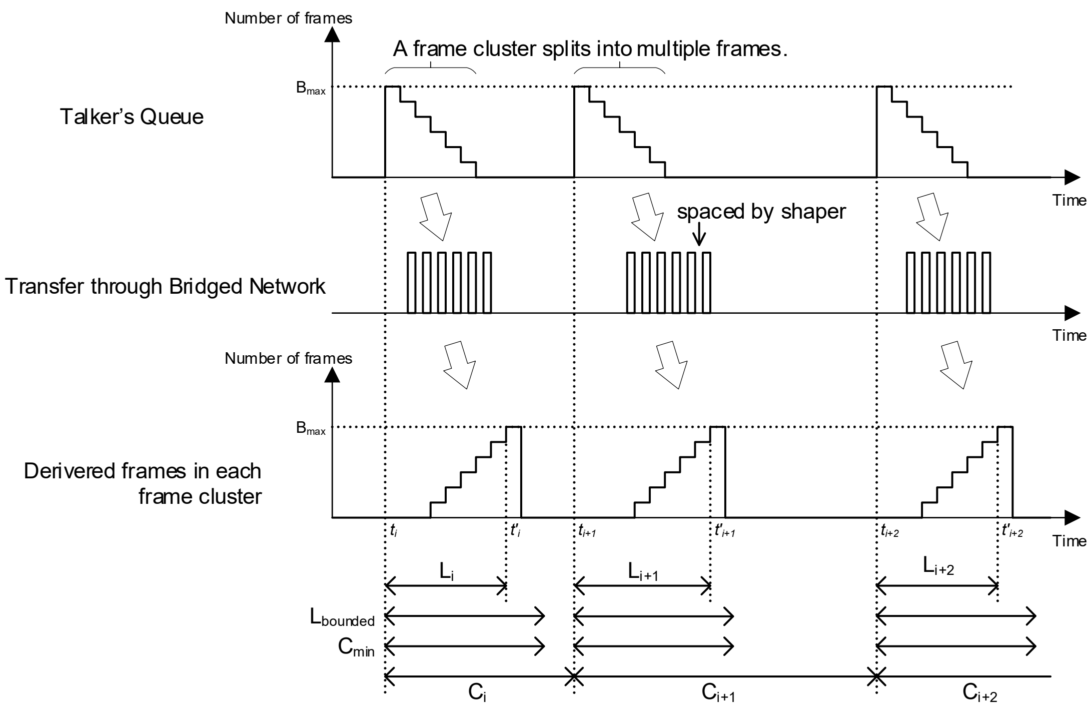

Figure X-4—Traffic pattern from the application's point of view

L_i represents $t'_i - t_i$, which is the time it takes the whole i-th frame cluster to be transferred from the queue of the Talker to the input queue of Listener. C_i equals to $t_{i+1} - t_i$. Both L_i and C_i often vary according to i. The reference points are assumed to be placed in the beginning of the frames in Figure X-4.

In the bridged network's point of view, an application's transaction might be transferred as multiple frames through the bridged network since the size of the transaction is typically larger than the maximum size of a frame the bridged network can handle. When the application requires bounded latency for the transaction, this causes bursty transmission of these frames and can result in disruption of other communication in the bridged network. To reduce it, shaper parameter settings for multiple frames are discussed next.

Bursty traffic is shaped by the Talker. The shaping rate the Talker uses is denoted by *S*. As a result of traffic shaping, the interval in which the Talker sends each frame becomes equal to the frame length divided by the shaping rate. Then, at the input of a Listener, the delivery time of this frame cluster (as shown in Figure X-4) becomes as follows:

$$t'_i = t_i + L_i = t_i + \frac{\sum_{k=1}^{n_i - 1} F_{i,k}}{S} + A_{i,n_i} \qquad \text{(X-1)}$$

where $A_{i,k}$ denotes latency experienced by the *k*-th frame of the *i*-th frame cluster when traveling from Talker to Listener. *S*, within the traffic shaper, is set so that the delivery time is within the bounded latency required by the application.

$F_{i,k}$ should be configured carefully by the system implementor because they affect other traffic in the network and depend on the parameters of the network.

X.3 Latency through Bridged Network

X.3.1 Network latency

Latency between Talker and Listener (hereinafter referred to as network latency) is derived from the sum of per-hop latency along the path between them. Network latency is affected by other traffic, internal processing of bridges, wire propagation time and media access delay and also varies per transfer of each frame. The maximum network latency should be met for bursty traffic requiring bounded latency. $A_{i,k}$ denotes the network latency of the *k*-th frame of the *i*-th frame cluster. Accordingly, A_{max} hereinafter denotes the maximum network latency of the frames of the focused traffic, that is, for all *i* and *k*.

A_{max} is also the upper bound of A_{i,n_i}. A_{max} is used as an input of the calculating procedure for shaper parameter settings described in X.4.

Estimation of A_{max} depends on the selection of mechanisms used in the forwarding process on bridges and Talker transmission process on end stations.

X.3.2 Credit-based shaper algorithm

Per-hop latency imposed by credit-based shaper algorithm (8.6.8.2) against a single frame is discussed in Annex L. Network latency is derived from the sum of per-hop latencies along the path.

IEEE Std 802.1BA [B9] also provides a method of latency calculation.

X.3.3 Static Reservation

A network administrator can configure streams reserved statically by setting up parameters of the forwarding process of the bridges and end stations in the network, for instance, egress filtering, flow classification, transmission selection and stream gate control. The administrator is responsible to calculation of A_{max} for each stream.

X.3.4 Dynamic Reservation by SRP

In bridged networks controlled by SRP, the value *portTcMaxLatency* and *AccumulatedLatency*[1] can be obtained from the system, while it can be manually evaluated in the same manner on networks without SRP. Both of them are evaluated on the worst-case scenario, that is, *AccumulatedLatency* is equivalent to A_{max}

MaxLatency is a parameter in UserToNetworkRequirements TLV of MSRPv1, which is related to each stream and defined as the upper bound of *AccumulatedLatency*. The value of *MaxLatency* is recommended to be determined by shaper parameter setting calculation or the network administrator as it is equal to or greater than *AccumulatedLatency*.

The calculation of efficient shaper parameter settings requires the accurate value of *AccumulatedLatency*, while *AccumulatedLatency* is the result of the reservation made with the calculated shaper parameter settings.

A process establishing a reservation often consists of multiple steps. A common case is that a path is searched first and then that some calculations and configurations are made according to the path. As mentioned in the previous clause, Accumulated Latency depends on a path from Talker to Listener, therefore, it is obtained after the path is determined. Of course, it is obtained prior to calculation.

In case of SRP (Clause 35), shaper parameter setting calculation utilizes SRP. *AccumulatedLatency* is obtained from SRP and TSpec is passed to SRP for reservation. To obtain *AccumulatedLatency*, firstly a Talker Advertise message is propagated along the paths in the network and evaluates the availability of the stream. Once it reaches a Listener, the corresponding Listener Ready message confirms the reservation. The Listener Ready message also carries information of the path to the Talker, including Accumulated Latency if all the nodes along the path support SRP version 1. The shaper parameter setting calculation may use the MaxLatency element of the UserToNetworkRequirements group (Clause 46.2.3.6.2) and the *AccumulatedLatency* group (Clause 46.2.5.2) in order to obtain *AccumulatedLatency* from the Talker. The SRP specification requires the Talker to send a Talker Advertise message and to collect Listener Ready message. Once a Listener Ready message reaches Talker, the path of the reservation is determined and *AccumulatedLatency* is reported. However, SRP is not capable of updating the parameters of the reservation confirmed by the Listener Ready message. To work with SRP, the reservation is dropped first and then a new reservation with new parameters is requested. The tentative parameters of the first Talker Advertise are derived assuming the *AccumulatedLatency* which can be set by implementer's choice, such as determining by the network administrator, and adopting a value of zero as simple recommendation. Then the second reservation is requested again with the amount obtained by the first request. The first calculated reservation and the second one is not guaranteed to return the same values of the *AccumulatedLatency*, therefore attempts to join with different parameters and *MaxLatency* based on the previously obtained *AccumulatedLatency* are repeatedly made until successful joining the target stream.

X.3.5 Enhancements for scheduled traffic

Enhancements for scheduled traffic (8.6.8.4) uses gates to stop sending out frames. Therefore, in case that this method is used, latency from Talker to Listener should increase by the maximum time while the gate consecutively closes. TimeInterval (8.6.9.4.23) is larger than the time while the whole of a frame is sent out.

[1] Calculation of these values is non-trivial and beyond this standard.

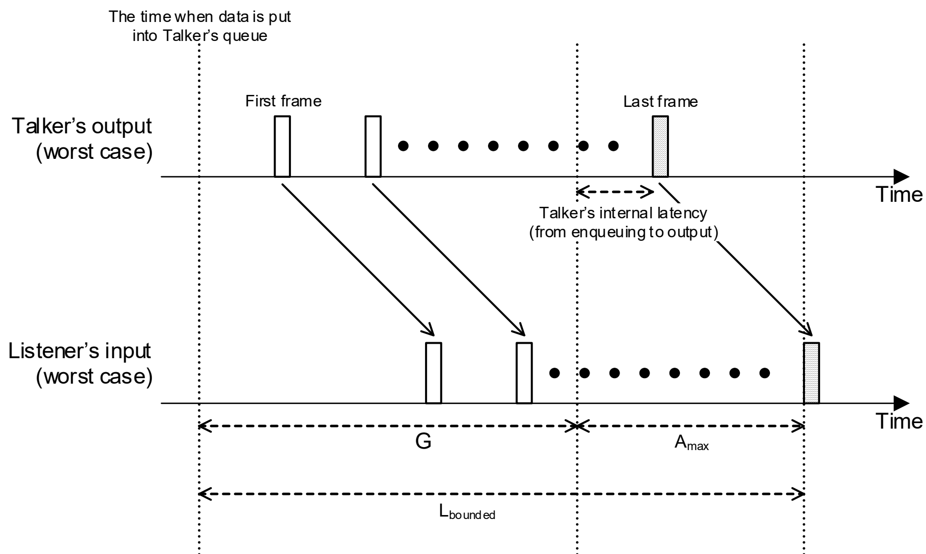

Figure X-5—The worst-case timing of the last frame

X.4 Shaper Parameter Settings

X.4.1 General Discussion of Shaping Rate

This standard defines several types of shapers. Each shaper causes intervals between frames, however the sizes of the intervals between the frames vary according to the type of the shaper and the parameters used to configure the shaper. Each shaper is discussed in the following subclauses.

In order to reduce over-provisioning of bandwidth reservation while providing the specified bounded latency, the bursty traffic should be shaped with the minimum shaping rate (hereinafter referred to as *R* and expressed in bits per second). Figure X-5 illustrates worst-case propagation of the last frame of a cluster comprising *n* frames within the given bounded latency while reducing over-provision of bandwidth reservation. Latency Budget (hereinafter referred to as *G* and expressed in seconds) is defined as the maximum duration while Talker emits (n-1) frames. Figure X-6 shows *G* can be derived from bounded latency and A_{max} as follow:

$$G = L_{bounded} - A_{max} \tag{X-2}$$

where the units of $L_{bounded}$ and A_{max} are seconds.

The minimum shaping rate for traffic shaping is equal to:

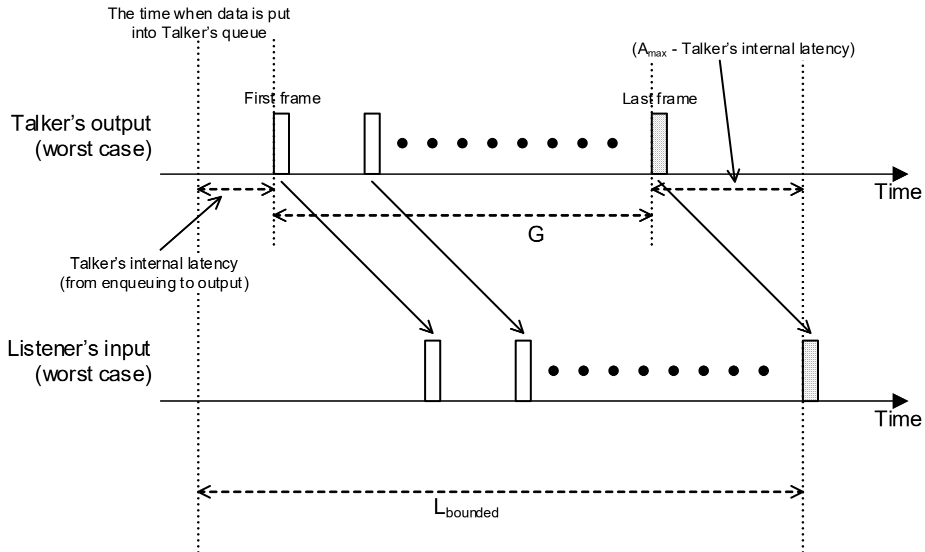

Figure X-6—The worst-case budget calculation

$$R = \frac{\sum_{k=1}^{n_{i_{worst}}-1} F_{i_{worst},k}}{G} = \frac{B_{i_{worst}} - F_{i_{worst}, n_{i_{worst}}}}{L_{bounded} - A_{max}} \tag{X-3}$$

where the units of B_i and $F_{j,k}$ are bits.

B_i is much larger than the length of the last frame (i.e. $FrameLength(i_{worst})$), therefore R can be simplified by introducing a small positive value ε as follows;

$$R = \frac{B_{i_{worst}}}{L_{bounded} - A_{max}} - \varepsilon \tag{X-4}$$

ε can be zero in actual implementations.

In case that an application developer requests to provide bounded latency until the last bit of the frame cluster delivers, consider the (n+1)th frame and its reference point that are imaginary. R is given to make the time between reference points of the first frame and this imaginary frame provide the bounded latency. Therefore, replacement $n_{i_{worst}}$ with $n_{i_{worst}} + 1$ in (X-3) results in (X-5).

$$R = \frac{\sum_{k=1}^{(n_{i_{worst}}+1)-1} F_{i_{worst}, k}}{G} \tag{X-5}$$

$$= \frac{B_{i_{worst}}}{L_{bounded} - A_{max}} = \frac{B_{max}}{L_{bounded} - A_{max}}$$

Talker needs to have memory buffer as large as the data size.

X.4.2 Credit-Based Shaper

idleSlope is the only parameter describing a credit-based shaper (8.6.8.2). The following equation follows from the equation X-6.

$$\text{idleSlop} = R \tag{X-6}$$

Both *idleSlope* and *R* are expressed in bits per second.

X.4.3 Asynchronous Traffic Shaping

According to the definition of the ATS scheduler state machine in Clause 8.6.11, *CommittedBurstSize* in bits (8.6.11.3.5) should be equal to or greater than frames sent by the Talker. In this case, it is recommended to be equal to the *Maximum SDU (Service Data Unit) Size*, which is defined in Clause 6.5.8. In this annex *Maximum SDU Size* is expressed in bits, while it is expressed in octets in other parts of this standard. *CommittedInformationRate* in bits per second (8.6.11.3.6) is the data rate reserved for the stream and is recommended to be equal to the *R* shown in Equation (X-3). The approximation discussed in Clause X.4.1 can also be applied. These lead to the following settings values:

$$\text{CommittedBurstSize} = \text{Maximum SDU Size} \tag{X-7}$$

$$\text{CommittedInformationRate} = \frac{B_{max}}{L_{bounded} - A_{max}} \tag{X-8}$$

Since the ATS scheduler state machine operation (8.6.11) assumes that the frame sizes that are processed are less than or equal to the associated *CommittedBurstSize* parameter (8.6.11.3.5), the *CommittedBurstSize* is set to be the maximum size of frame. That is equal to the *Maximum SDU Size* as shown in Equation (X-7).

X.4.4 Traffic Specification in SRP

The MSRP TSpec is used in the credit-based shaper transmission selection algorithm. This type of TSpec is intended for use by reservations that compatibly supports AVB SR class A or SR class B. Unlike audio/video streaming, TSpec for bursty traffic, which characterizes the bandwidth that a stream can consume, needs to consider *MaximumDataSize* and $L_{bounded}$.

The TSpec parameters for MSRP are recommended to be set as follows:

$$\mathrm{MaxFrameSize} = \min\left(\left\lfloor \frac{B_{\mathrm{max}}}{L_{\mathrm{bounded}} - A_{\mathrm{max}}} \times \mathrm{classMeasurementInterval} \right\rfloor, \text{Maximum SDU Size}\right) \tag{X-9}$$

$$\mathrm{MaxIntervalFrame} = \left\lceil \frac{1}{\mathrm{MaxFrameSize}} \times \frac{B_{\mathrm{max}}}{L_{\mathrm{bounded}} - A_{\mathrm{max}}} \times \mathrm{classMeasurementInterval} \right\rceil \tag{X-10}$$

The *Maximum SDU (Service Data Unit) size* is defined in (6.5.8) and is expressed in bits here. *classMeasurementInterval* is defined in (34.3) and expressed in seconds. The *MaxFrameSize* is recommended to set the *Maximum SDU Size*. However, the *MaxFrameSize* should be smaller than the *Maximum SDU size* in case that *classMeasurementInterval* is shorter, i.e. the number of bytes within the *classMeasurementInterval* is smaller than the *Maximum SDU size*. The *MaxIntervalFrame* needs to be guaranteed so as to become a positive integer (1 or larger value).

When considering the definition of the value of the TrafficSpecification TLV as defined in (35.2.2.10.6) and TrafficSpecification elements of User/network configuration information defined in (46.2.3.5), then Equation (X-9) and Equation (X-10) can be presented as follows:

$$\mathrm{MaxFrameSize} = \min\left(\left\lfloor \frac{B_{\mathrm{max}}}{L_{\mathrm{bounded}} - A_{\mathrm{max}}} \times \mathrm{Interval} \right\rfloor, \text{Maximum SDU Size}\right) \tag{X-11}$$

$$\mathrm{MaxFramesPerInterval} = \left| \frac{1}{\mathrm{MaxFrameSize}} \times \frac{B_{\mathrm{max}}}{L_{\mathrm{bounded}} - A_{\mathrm{max}}} \times \mathrm{Interval} \right| \tag{X-12}$$

The parameter Interval is referred in (46.2.3.5.1), which replaced *classMeasurementInterval* in Equation (X-9) and Equation (X-10). The unit of *Interval* is seconds. The *Interval* is recommended to be set less than the bounded latency for controlling the shaping rate during the shaping duration.

X.4.5 Use of enhancements for scheduled traffic

Enhancements for scheduled traffic mechanism works as the intended shaper by configuring its gate control list in which the number of “Open” equals to *NumOfOpen* derived by the following equation:

$$\mathrm{NumOfOpen} = \left\lceil \frac{\max n_i}{\left\lfloor \frac{\mathrm{TimeInterval} \times \mathrm{portTransmitRate}}{\max F_{i,k}} \right\rfloor \times \left\lfloor \frac{L_{\mathrm{bounded}}}{\mathrm{OperCycleTime}} - 1 \right\rfloor} \right\rceil \tag{X-13}$$

portTransmitRate is defined in (8.6.8.2) and expressed in bits per second, and that the unit of *TimeInterval* above is seconds, while *TimeInterval* defined in (8.6.9.4.23) is practically expressed in nanoseconds.

Annex Y

(informative)

Bibliography

The following text will not be changed, but has been included for the benefit of developers and reviewers of this draft standard.

Bibliographical references are resources that provide additional or helpful material but do not need to be understood or used to implement this standard. Reference to these resources is made for informational use only.

Insert the following references, renumbering as necessary.

[B1] IEEE 802 Nendica Report: Flexible Factory IoT: Use Cases and Communication Requirements for Wired and Wireless Bridged Networks, IEEE-SA Industry Connections, Apr. 2020, https://ieeexplore.ieee.org/servlet/opac?punumber=9068509 [2,3]

[2] IEEE publications are available from The Institute of Electrical and Electronics Engineers (https://standards.ieee.org/).

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